# SQL DB Geo Replication and Data Movement to Blob Storage

## Objective

The objective of this exercise is to understand how data can be moved from an OLTP SQL database to Azure Blob Storage for analytics purposes. In this lab, we learn how to configure geo-replication, create a storage account with cool tier, and use Azure Data Factory to copy data.

## Introduction

In real-world applications, OLTP databases are mainly used for transactional operations. They are not suitable for heavy analytical queries. So, organizations usually replicate data to another region and move it to blob storage for further analysis.

In this exercise, I learned how Azure provides geo-replication and data movement services to handle such scenarios efficiently.

## Step 1 – Creating Azure SQL Database

First, an Azure SQL Database is created using the AdventureWorks sample database.

This database acts as the primary database and contains sales-related tables such as SalesOrderDetail.

At this stage, I understood how SQL Database in Azure works as a managed database service.

## Step 2 – Configuring Active Geo-Replication

Next, Active Geo-Replication is configured.

The SQL Database is replicated to another Azure region (for example: Central India or South East Asia).

The secondary database is created as a readable replica.

This helps in reducing load on the primary database and provides high availability.

From this step, I understood that geo-replication improves reliability and disaster recovery.

## Step 3 – Creating Azure Blob Storage (Cool Tier)

A new Storage Account is created.

While creating the storage account:

- Access tier is selected as **Cool**.
- This helps in reducing storage cost for less frequently accessed data.

Blob Storage will act as the destination where data from SQL DB will be copied.

## Step 4 – Creating Azure Data Factory

Azure Data Factory (ADF) is created to perform data movement.

Inside Data Factory:

1. A new pipeline is created.
2. Source is selected as the readable SQL DB replica.

3. Destination is selected as Azure Blob Storage.
4. SalesOrderDetail table is selected as source table.

This pipeline copies data from SQL database to blob storage.

I understood that Data Factory is mainly used for ETL (Extract, Transform, Load) operations.

## Step 5 – Running the Pipeline

After configuring the pipeline, it is triggered manually.

The pipeline copies the data successfully from SQL DB to Blob Storage.

## Step 6 – Verifying Data in Blob Storage

Finally, the Blob container is opened to verify if the sales data file is available.

If the file is present, it confirms successful data transfer.

This step ensures that data movement is completed correctly.

## Regions Used

The following Azure regions can be used:

- South India
- Central India
- South East Asia

## Conclusion

From this exercise, I understood how Azure SQL Database supports geo-replication for high availability. I also learned how Azure Blob Storage can be used to store analytical data in cool tier for cost optimization. Using Azure Data Factory, data can be copied from OLTP systems to storage for analytics.

This exercise helped me understand real-time enterprise data movement concepts in Azure.